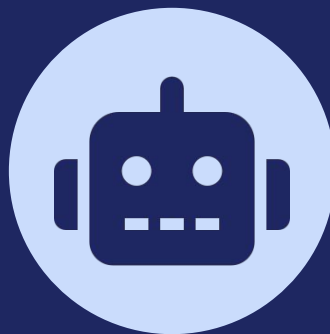




AI & Machine Learning

Understanding the Basics

AWS Certified Machine Learning Associate • Part 1

Presented by Prajan Shrestha

What We'll Cover



What is Artificial Intelligence?



What is Machine Learning?



How AI, ML, and Deep Learning relate



How machine learning actually learns



The main types of machine learning



Real-world applications & where things stand today

What is Artificial Intelligence?

AI is the broad field of building machines that can perform tasks that normally require human intelligence.

That includes things like understanding language, recognizing images, making decisions, and solving problems.

AI isn't one single technology. It's an umbrella term covering everything from simple rule-based systems to today's most advanced models.



AI is already part of daily life, often without us noticing.



Voice assistants — Siri, Alexa



Self-driving vehicles



Chatbots & virtual agents

What is Machine Learning?



Machine learning (ML) is one of the main ways we build AI today.

Instead of programming explicit rules for every situation, we give a computer data and let it find patterns on its own. The system learns from examples and improves over time without being told exactly what to do for every case.

Traditional Programming

Rules + Data



Answers

A human writes every rule by hand

Machine Learning

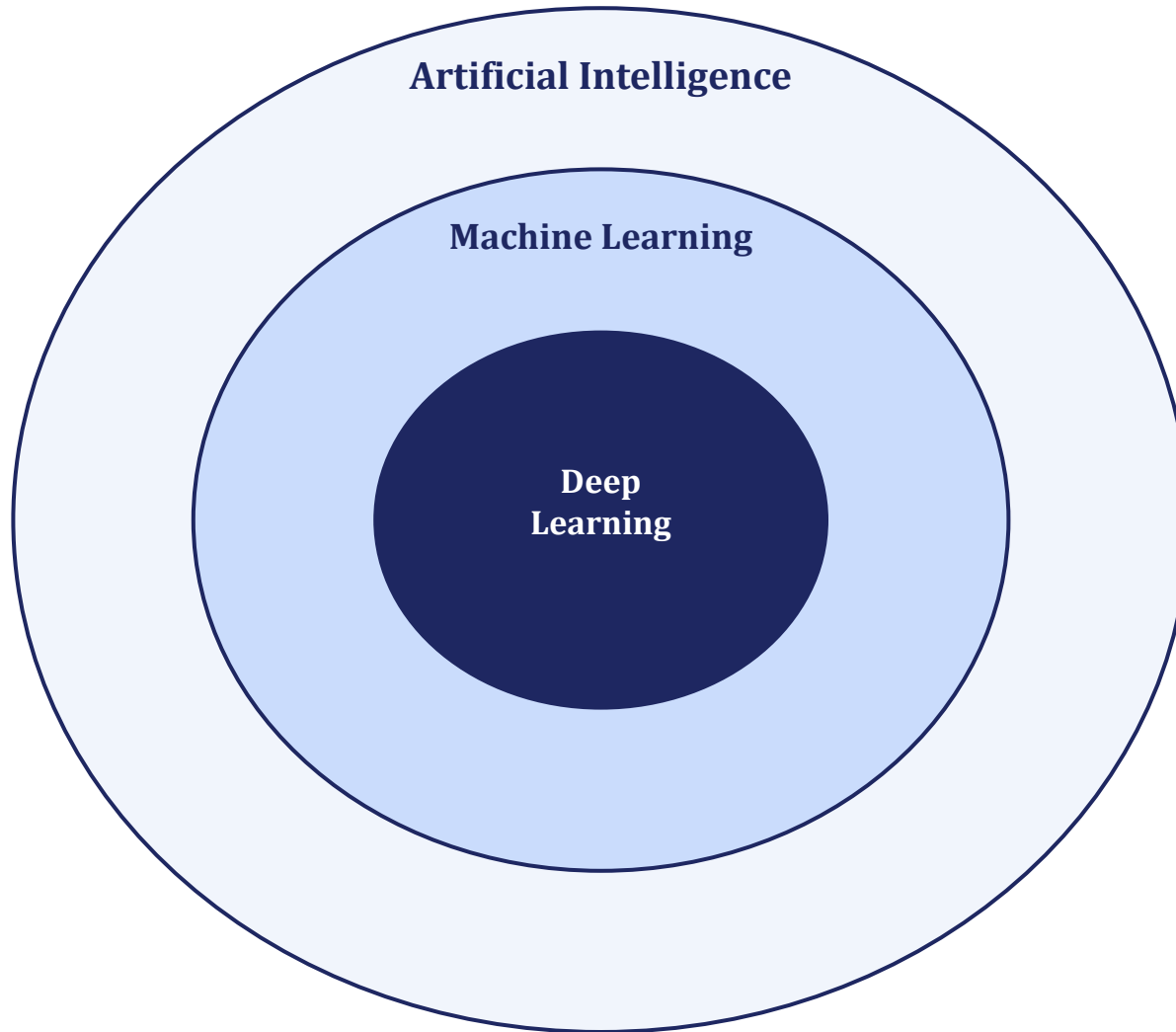
Data + Answers



Rules (the model)

The system works out the rules itself

How AI, ML, and Deep Learning Fit Together



Artificial Intelligence

The overall goal: machines that act intelligently

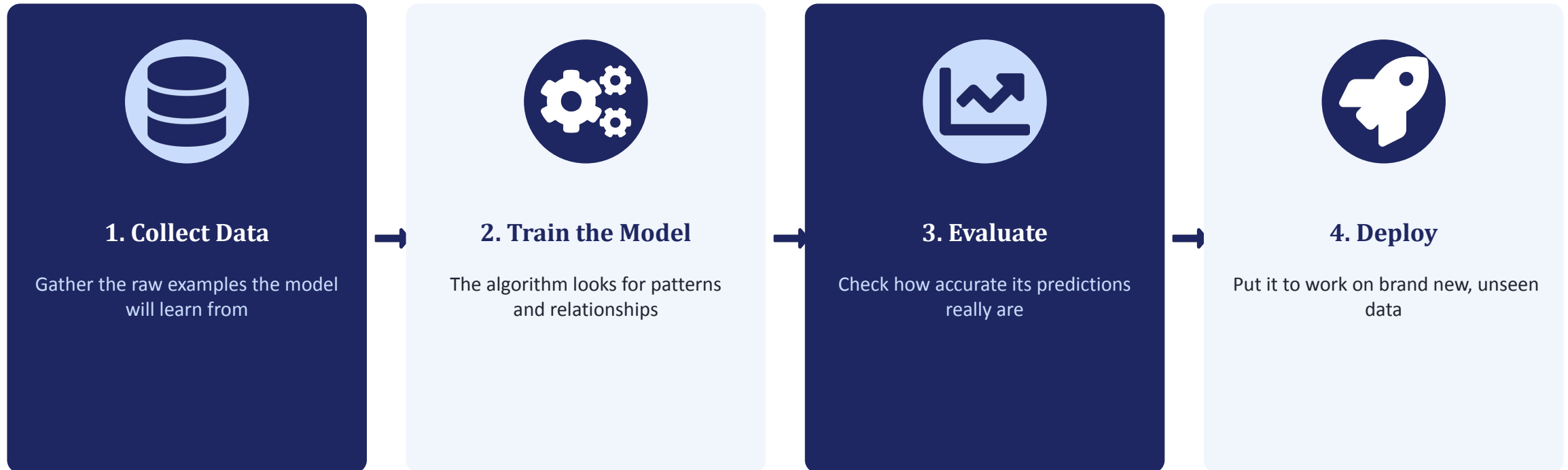
Machine Learning

A data-driven approach to reach that goal

Deep Learning

Layered neural networks for complex problems

How Does a Machine Learning System Learn?



This cycle repeats: models are retrained as new data comes in.

Three Main Types of Machine Learning



Supervised Learning

Learns from labeled data, where the correct answer is already known.

e.g. predicting prices, spam detection



Unsupervised Learning

Finds hidden patterns or groupings in unlabeled data on its own.

e.g. customer segmentation



Reinforcement Learning

Learns through trial and error, earning rewards for good decisions.

e.g. game-playing agents, robotics

Where You Already See Machine Learning



Healthcare

Earlier disease detection from medical scans



Finance

Real-time fraud detection



Retail

Personalized product recommendations



Transportation

Route optimization & self-driving tech



Entertainment

Recommending your next show



Manufacturing

Predicting maintenance before failure

Where Things Stand Right Now



Generative AI

Large language models behind tools like ChatGPT and Claude have pushed machine learning into the mainstream.

These systems can now:

- Write and summarize text
- Generate images and audio
- Write and explain code
- Hold natural conversations

The global AI market is growing rapidly, and companies across every industry are racing to build ML into their products.

Why This Matters for AWS



AWS gives developers and businesses the infrastructure to build, train, and deploy ML models without managing physical hardware or building everything from scratch.

Amazon SageMaker

Handles the ML lifecycle we just covered, end to end:

- Prepare & label data
- Build & train models
- Tune & evaluate performance
- Deploy models at scale

That's exactly what this course dives into over the coming weeks.

Recap

- AI is the big-picture goal: building intelligent systems.
- ML is the data-driven approach we use to get there.
- It's already reshaping healthcare, finance, retail, and more.

Thanks for watching!